# EECS 498/CSE 598: Zero-Knowledge Proofs
# Winter 2026
# Lecture 10

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# Agenda for this lecture

- Announcements
- Compiling PIOPs to arguments/Sigma protocols
- Schnorr’s protocol for discrete logarithms
- Okamoto’s protocol for Pedersen commitments
- Other sigma protocols

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# Announcements

- Project 2 is online
- Autograder is working
- Due date: 2/20
- PAG travelling 3/9 3/11

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Arg[PIOP, PC]

# Compiling PIOPs to arguments

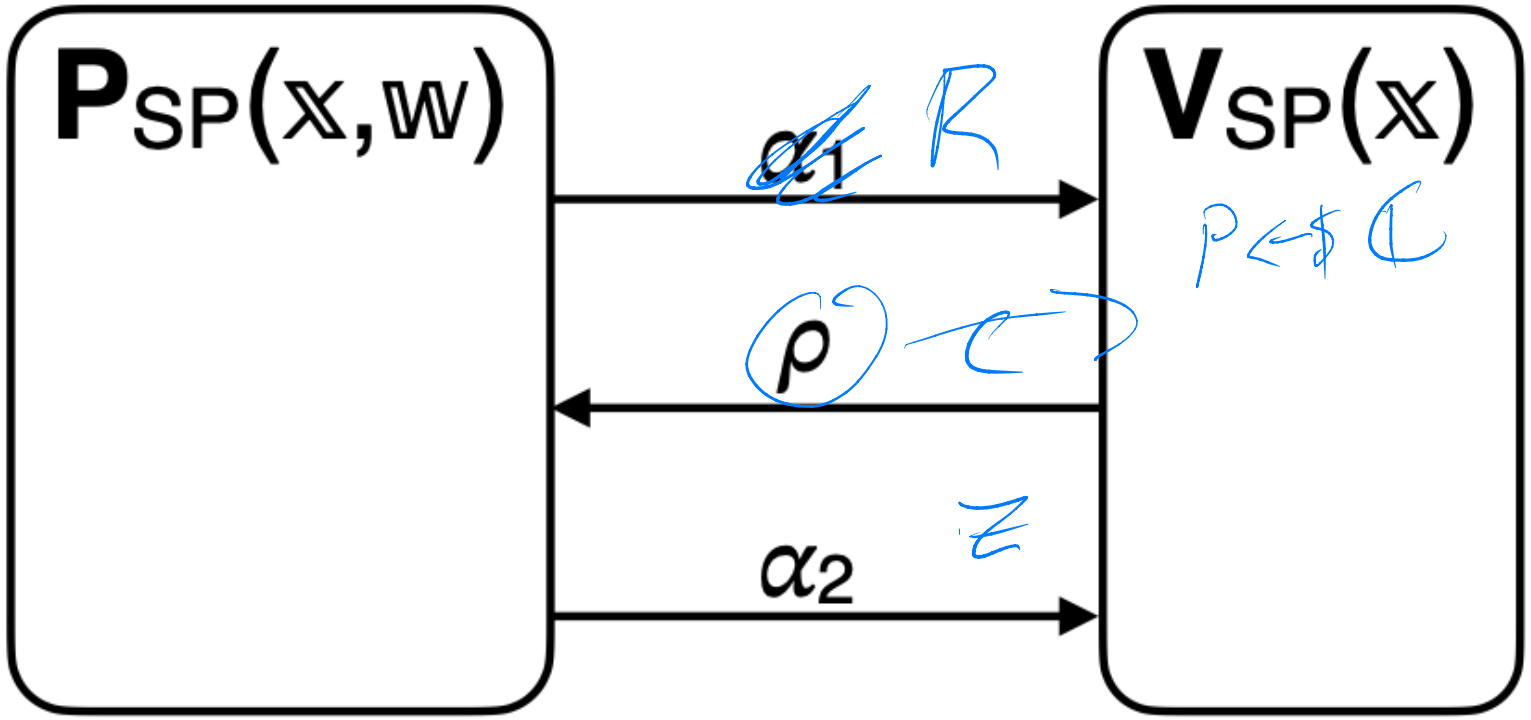
- Whenever $\mathbb{P}$ sends $\mathcal{O}_p$, the Arg[PIOP, PC] comm. fs to $\mathbb{P}$ using PC and sends comm to $\mathbb{V}$
- On query $r$ to $\mathcal{O}_p$ from $\mathbb{V}_{PIOP}$, Arg[PIOP, PC].$\mathbb{V}$ do open(comm, $r$, $v$)
- Arg $\mathbb{V}$ outputs whatever PIOP $\mathbb{V}$ outputs

# Compiling PIOPs to arguments

**Claim.** *Let* $\mathsf{PIOP} = \langle \mathcal{P}_{\mathsf{PIOP}}, \mathcal{V}_{\mathsf{PIOP}} \rangle$ *be a polynomial interactive oracle proof for* $\mathcal{L}$ *with degree and variable bounds* $(d_1, v_1), \ldots, (d_k, v_k)$. *Let* $\mathsf{PC} = (\mathsf{Setup}, \mathsf{Com}, \mathsf{Open})$ *be a polynomial commitment scheme. Let* $\mathsf{Arg}[\mathsf{PIOP}, \mathsf{PC}]$ *be the argument obtained from the "oracle-to-PC" compiler. If* $\mathsf{PC.Com}$ *is binding,* $\mathsf{PC.Open}$ *is a proof of knowledge, and* $\mathsf{PIOP}$ *is sound, then* $\mathsf{Arg}[\mathsf{PIOP}, \mathsf{PC}]$ *is a sound argument for* $\mathcal{L}$.

# Sigma protocols

"Zig"
MA
Merlin-Arthur




$\rho \leftarrow\$ \mathbb{C}$

V

verify$(\mathbb{x}, \alpha_1, \rho, \alpha_2)$
$\rightarrow \{0,1\}$

# Security for sigma protocols

Let $SP = \langle P, V \rangle$ be a $\Sigma$-protocol for $\mathcal{R}$

$SP$ is $k$-SS if $\exists$ determ. Ext

s.t. whenever Ext is given an input $x$

and $k$ accepting transcripts on same $R$

$c_i$ distinct

$$\underbrace{(R, c_1, z_1) \ldots (R, c_k, z_k)}$$

$$\mathrm{Ext}(x, \quad) = w \quad \text{st } (x, w) \in \mathcal{R}$$

$\boxed{\text{Special HVZK}} \rightarrow$ next week

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# Schnorr's protocol

| $\mathsf{Schnorr}[\mathcal{P}](\mathbb{G}, G, p, X; x)$ | | $\mathsf{Schnorr}[\mathcal{V}](\mathbb{G}, G, p, X)$ |
|---|---|---|
| $r \leftarrow\$ \mathbb{Z}_p \; ; \; R \leftarrow rG$ | | |
| | $\xrightarrow{\quad R \quad}$ | |
| | | $c \leftarrow\$ \mathbb{Z}_p$ |
| | $\xleftarrow{\quad c \quad}$ | |
| $z \leftarrow r + cx$ | | |
| | $\xrightarrow{\quad z \quad}$ | |
| | | Return $zG = R + cX$ |

$$Z = r + cX$$

$$ZG = rG + cXG$$

$$= rG + cXG \rightarrow cX$$

# Completeness and 2-special soundness

$\cancel{X}$ $(R, c_1, z_1)$

$G, P$ $(R, c_2, z_2)$

$$\begin{cases} z_1 G = R + c_1 \cancel{X} \\ z_2 G = R + c_2 \cancel{X} \end{cases}$$

$$(z_1 - z_2) G = (c_1 - c_2) \cancel{X}$$

$$\frac{z_1 - z_2}{c_1 - c_2} G = \cancel{X}$$

Schnorr is 2-SS $\boxtimes$

$$\frac{z_1 - z_2}{c_1 - c_2} = X$$

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# Okamoto's protocol

$C = aG + bH$

| $\mathsf{Okamoto}[\mathcal{P}](\mathbb{G}, G, H, p, C; a, b)$ | | $\mathsf{Okamoto}[\mathcal{V}](\mathbb{G}, G, H, p, C)$ |
|---|---|---|
| $\alpha, \beta \leftarrow\$ \mathbb{Z}_p \; ; \; R \leftarrow \alpha G + \beta H$ | | |
| | $\xrightarrow{\quad R \quad}$ | |
| | | $c \leftarrow\$ \mathbb{Z}_p$ |
| | $\xleftarrow{\quad c \quad}$ | |
| $z_a \leftarrow \alpha + ac$ | | |
| $z_b \leftarrow \beta + bc$ | | |
| | $\xrightarrow{\quad z_a, z_b \quad}$ | |
| | | Return $z_a G + z_b H = R + cC$ |

# Completeness and 2-SS for Okamoto

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# Chaum-Pedersen protocol for DDH

$\underline{\mathsf{DDH}_b^{\mathbb{G},G,p}(\mathcal{A})\text{:}}$

$a, b, c \leftarrow\$ \mathbb{Z}_p \; ; \; A \leftarrow aG \; ; \; B \leftarrow bG$

$C_0 \leftarrow abG \; ; \; C_1 \leftarrow cG$

$b' \leftarrow\$ \mathcal{A}(A, B, C_b, \mathbb{G}, G, p)$

Return $b'$

$$\mathsf{AdvDDH}^{\mathbb{G},G,p}(\mathcal{A}) = \left|\Pr[\mathsf{DDH}_0^{\mathbb{G},G,p}(\mathcal{A}) = 1] - \Pr[\mathsf{DDH}_1^{\mathbb{G},G,p}(\mathcal{A}) = 1]\right|$$

# Chaum-Pedersen protocol for DDH

| $\mathsf{ChPed}[\mathcal{P}](\mathbb{G}, G, A, B, C; b)$ | | $\mathsf{ChPed}[\mathcal{V}](\mathbb{G}, G, A, B, C)$ |
|---|---|---|
| $\beta \leftarrow\$ \mathbb{Z}_p$ | | |
| $R_G \leftarrow \beta G\ ;\ R_A \leftarrow \beta A$ | | |
| | $\xrightarrow{R_G, R_A}$ | |
| | | $c \leftarrow\$ \mathbb{Z}_p$ |
| | $\xleftarrow{c}$ | |
| $z \leftarrow \beta + cb$ | | |
| | $\xrightarrow{z}$ | |
| | | Return $zG = R_g + cB$ |
| | | $\wedge\ zA = R_A + cC$ |

$$\mathcal{R} = \{ (G, A, B, C; b) : B = bG \wedge C = bA \}$$

# Completeness and 2-SS for Chaum-Pedersen

$A, B, C$

$G$

$((R_G, R_A), c_1, z_1)$

$((R_G, R_A), c_2, z_2)$

$c_1 \neq c_2$

$z_1 G = R_G + c_1 B \iff z_2 G = R_G + c_2 B$

$z_1 A = R_A + c_1 C \iff z_2 A = R_A + c_2 C$

$$\frac{z_1 - z_2}{c_1 - c_2} G = B$$

$$\frac{z_1 - z_2}{c_1 - c_2} A = C$$